SINGAPORE DIMENSIONS • KINDERGARTEN (KA) • CHAPTER 1

# Numbers to 5: Counting & Cardinality

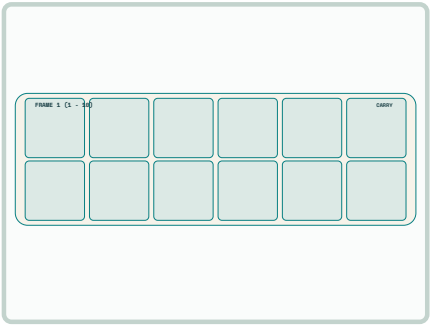
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.K.CC.B.4

Student Name: _____ Date: _____ **Model: SC-6** **Check Score: ___ / 4**

**STUDENT SELF-CHECK RULE (NO CHEATING):**
Build with blocks and solve on paper first! Use your SC-6 calculator ONLY to verify answers on the 4-line LCD screen.

## STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

### 1. 2D Bench Blueprint




Kit: Ten Frame Tray

### 2. Desk Setup (Concrete -> Pictorial)

**Physical Bench Setup:**
Place two-color counters in the top row of the 2x5 frame to represent 1 to 5.

**Sketch Model on Paper:**
Color in circles in a 5-frame grid on the worksheet.

### 3. Step-by-Step LCD Stack Trace (SC-6)

MODE: 123 Base-10 Arithmetic

| | | | |
|---|---|---|---|
| 1. | 4 | X: 4 | Key 1st operand |
| 2. | ENTER | Y: 4 \| X: 4 | Stack lift copies X |
| 3. | 1 | Y: 4 \| X: 1 | Key 2nd operand |
| 4. | + | X: 5 | Drop stack & compute |

EXPECTED LCD X: 5 counters

**1** **Place 2 counters on frame. Add 1 more counter.**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**2** **Place 3 counters on frame. Add 2 more counters.**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**3** **You have 4 counters. Remove 1 counter.**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

**4** **What number represents an empty ten-frame?**

Manipulative Model & Desk Sketch (Show your work):

**Final Math Answer:** _____

☐ Verified on SC-6

# Station Diagnostic & Solution Key: Chapter 1

Numbers to 5: Counting & Cardinality • Manipulative Setup, Diagnostic Rubric & Verification Flow

## MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

**Station Kit: Ten-Frame Studio (2x5 Grid) + Counters**

CAD Blueprint: ten_frame_tray

**Target Device: SC-6 Elementary**

**CAD / STL Source File:**

Education/scad/lib/ten_frame_tray.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

**Recommended Filaments:**

Oceanic Teal & Off-White (PLA)

| # | Math Problem Prompt | Expected Math Answer | Self-Check Keystrokes & RPN Stack |
|---|---|---|---|
| 1 | **Place 2 counters on frame. Add 1 more counte**<br>Standard: CCSS.MATH.K.CC.B.4 | 3 | [2] [ENTER] [1] [+] |
| 2 | **Place 3 counters on frame. Add 2 more counte**<br>Standard: CCSS.MATH.K.CC.B.4 | 5 | [3] [ENTER] [2] [+] |
| 3 | **You have 4 counters. Remove 1 counter.**<br>Standard: CCSS.MATH.K.CC.B.4 | 3 | [4] [ENTER] [1] [-] |
| 4 | **What number represents an empty ten-frame?**<br>Standard: CCSS.MATH.K.CC.B.4 | 0 | [0] or [C] |

## TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

**Key Misconception to Target During Station Observation:**

Watch for students recounting from 1 instead of continuing from the current register value.

**How the Manipulatives & Calculator Foster Self-Correction:**

Students solve with pencil and physical blocks first. When they self-check on the SC-6, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

**Station Facilitation & Pacing Tip:**

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

## CLASSROOM SCORING & STATION MASTERY RUBRIC

| | |
|---|---|
| **Exemplary (18–20 pts):** | Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack. |
| **Proficient (14–17 pts):** | Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting. |
| **Developing (10–13 pts):** | Understands problem concept but made computational slips; successfully self-corrected using calculator LCD. |
| **Intervention (<10 pts):** | Relied on calculator before solving on paper, or struggled with underlying physical manipulative models. |